

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

Процесс выполнения лабораторной работы

Работа с Midnight Commander

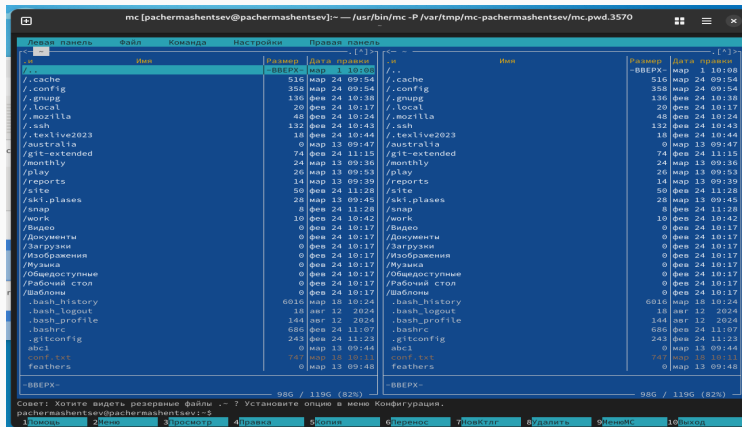


Рис. 1: Запуск mc

Работа с Midnight Commander

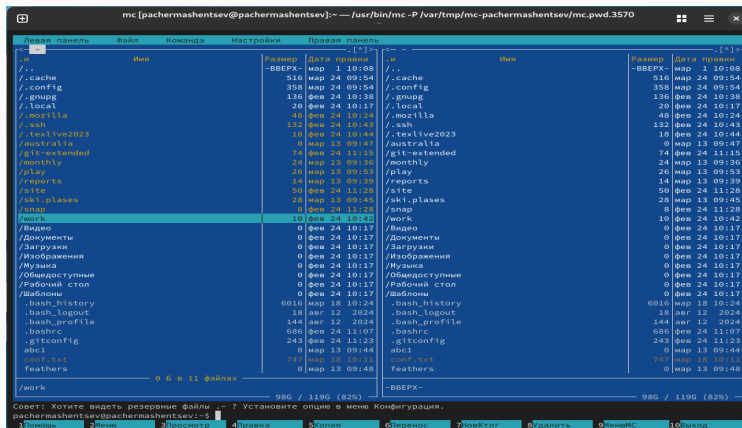


Рис. 2: Выделение

Работа с Midnight Commander

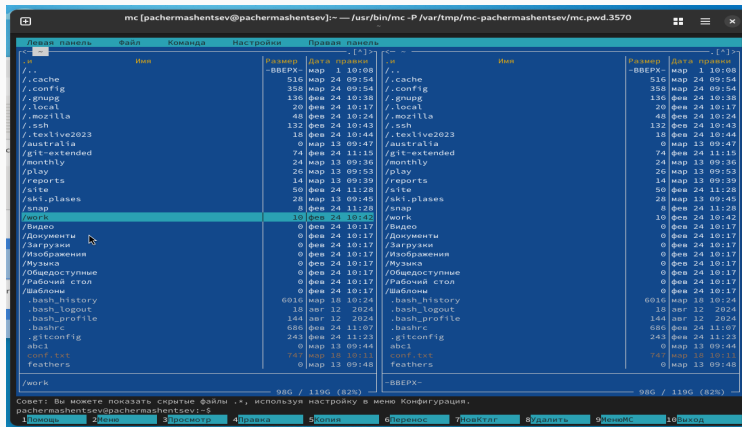


Рис. 3: Отмена

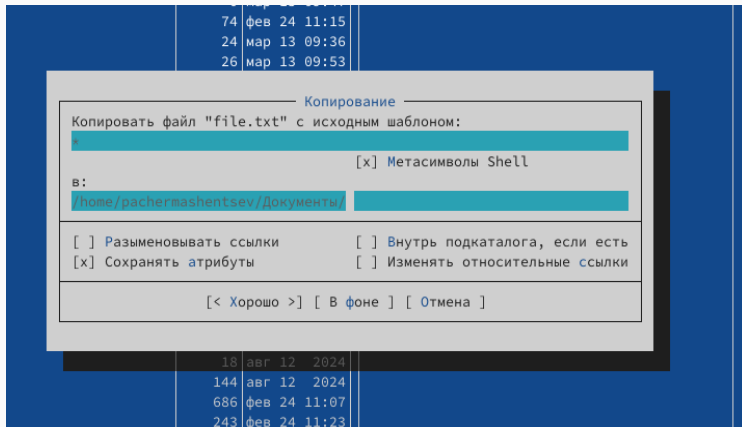


Рис. 4: Копирование

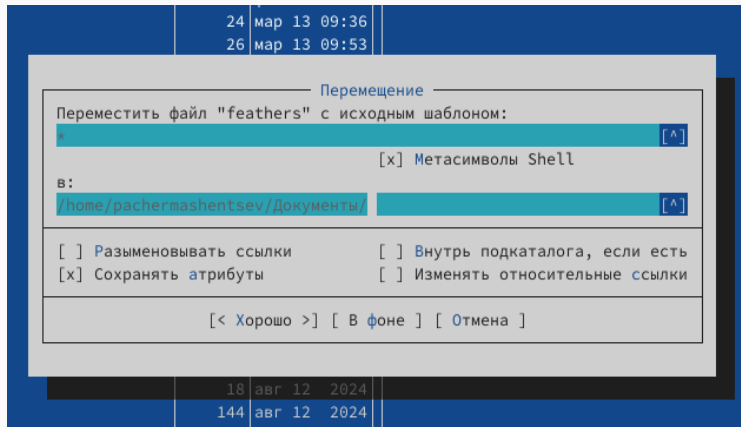


Рис. 5: Перемещение

Работа с Midnight Commander

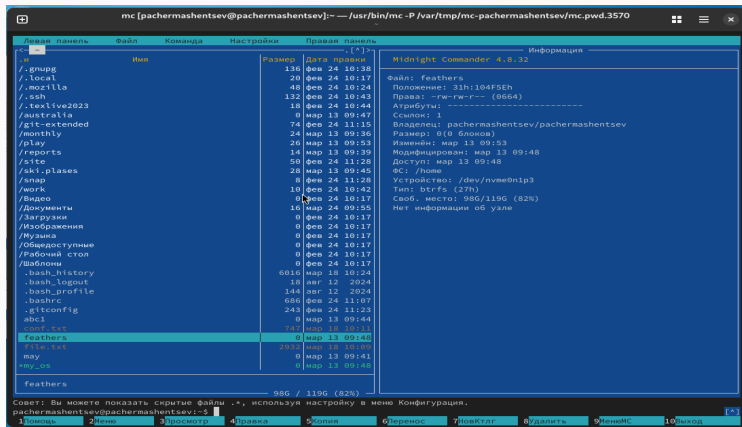


Рис. 6: Информация

Работа с Midnight Commander

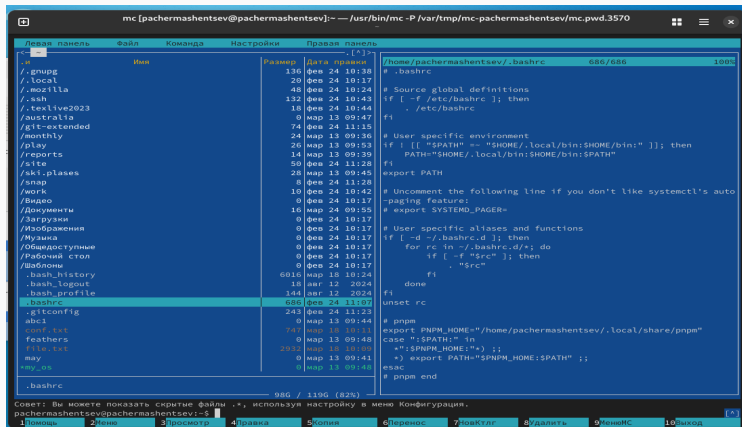


Рис. 7: Быстрый просмотр

Работа с Midnight Commander

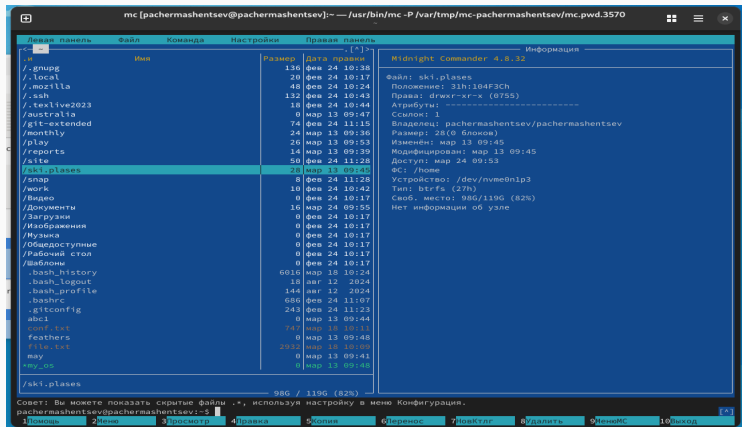


Рис. 8: Информация

Работа с Midnight Commander

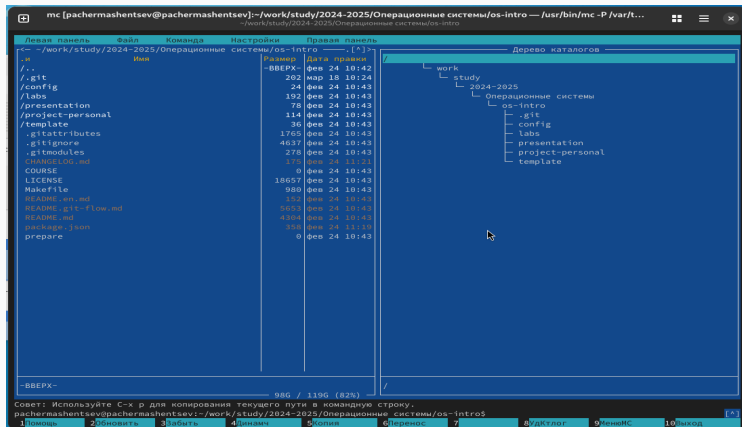
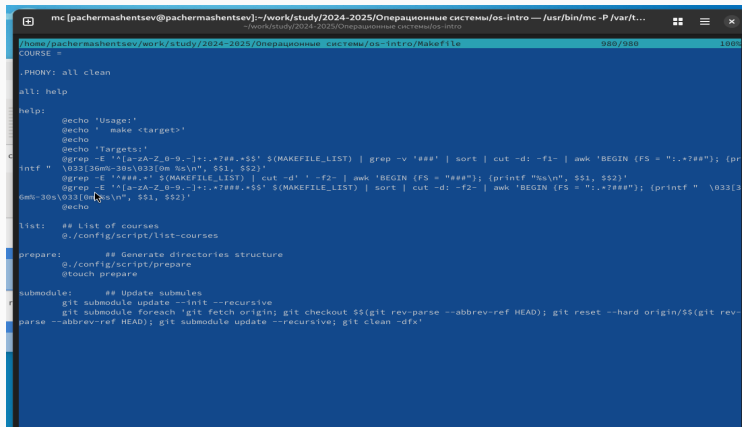


Рис. 9: Дерево каталогов



The screenshot shows the Midnight Commander interface. The title bar indicates the user is 'mc [pachermashentsev@pachermashentsev]'. The current directory is '/work/study/2024-2025/Операционные системы/os-intro'. The file being viewed is '/home/pachermashentsev/work/study/2024-2025/Операционные системы/os-intro/Makefile', which is 980/980 bytes and 100% text. The file content is a Makefile with the following targets and rules:

```
.PHONY: all clean

all: help

help:
    @echo 'Usage:'
    @echo '  make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ":.*?##"}; {pr
printf " \033[36m%-30s\033[0m %s\n", $1, $2}'
    @grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $1, $2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ":.*?##"}; {printf " \033[3
6m%-30s\033[0m %s\n", $1, $2}'
    @echo

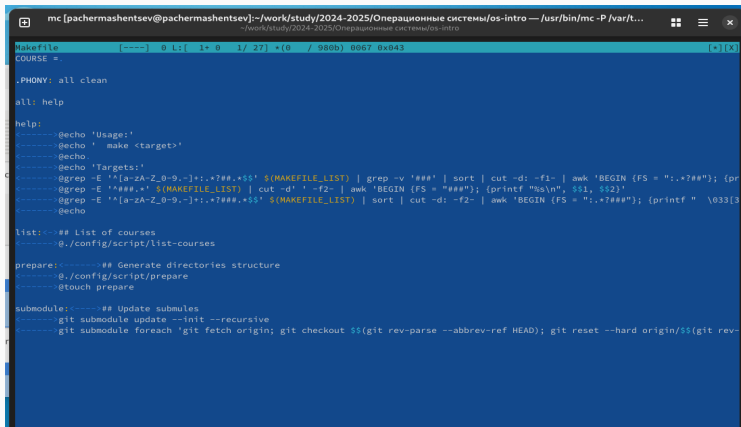
list: ## List of courses
    @./config/script/list-courses

prepare: ## Generate directories structure
    @./config/script/prepare
    @touch prepare

submodule: ## Update submodules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
parse --abbrev-ref HEAD); git submodule update --recursive; git clean -dfx'
```

Рис. 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



```
mc [pachermashentsev@pachermashentsev]~:/work/study/2024-2025/Операционные системы/os-intro — /usr/bin/mc -P /var/t...
~/.work/study/2024-2025/Операционные системы/os-intro
Makefile [-----] 0 L: [ 1+ 0 1/ 27] *(0 / 980b) 0067 0x043 [*)(X]
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
<----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $1, $2}'
<----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf " \033[3
<----->@echo

list:<-->## List of courses
<----->@./config/script/list-courses

prepare:<----->## Generate directories structure
<----->@./config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
```

Рис. 11: Отредактируем содержимое текстового файла без сохранения результатов

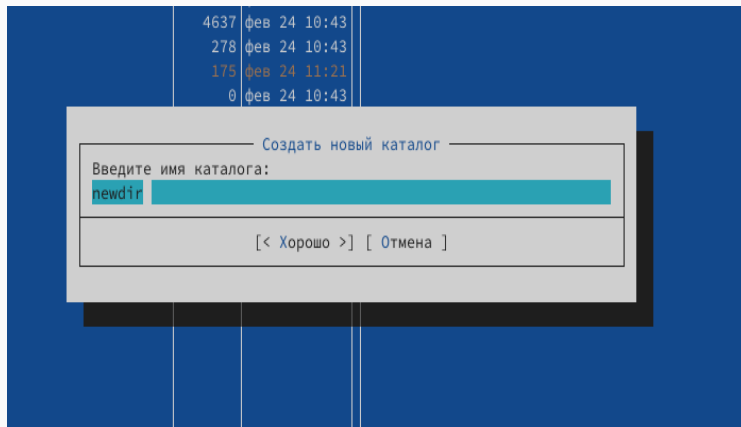


Рис. 12: Создание каталога

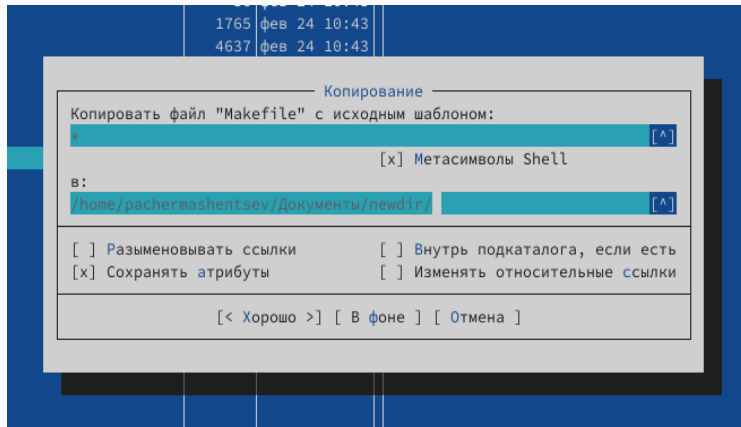


Рис. 13: Копирование в файлов в созданный каталог

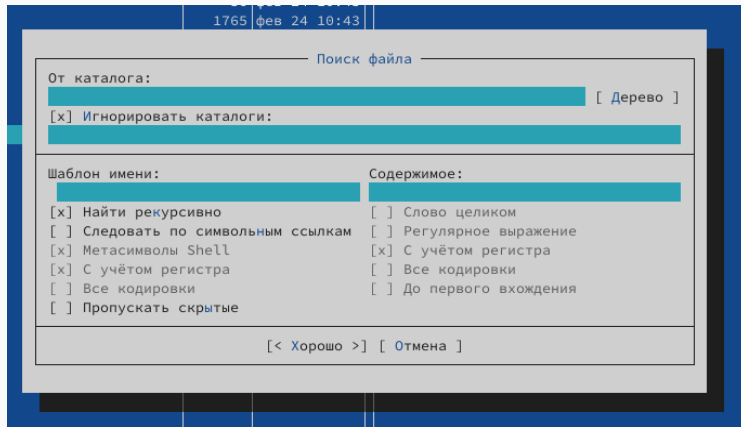


Рис. 14: Поиск файлов

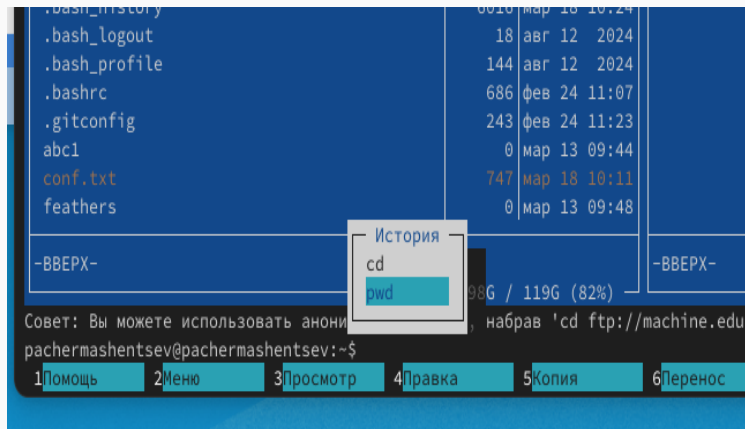


Рис. 15: История команд

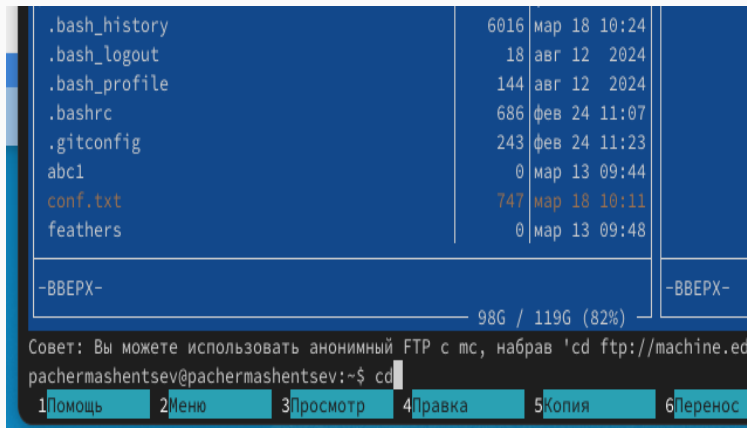
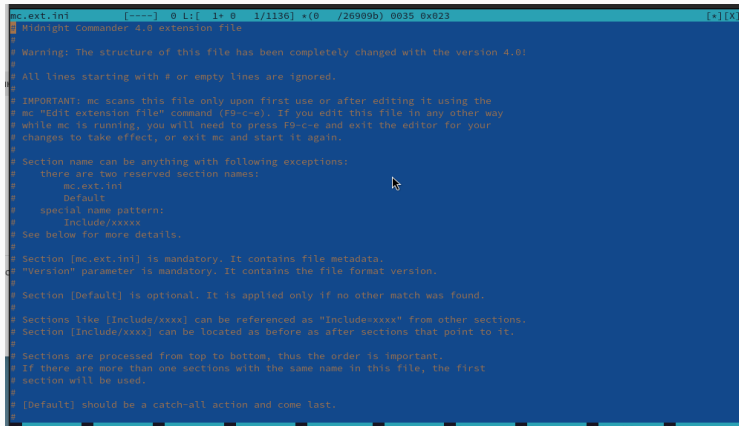


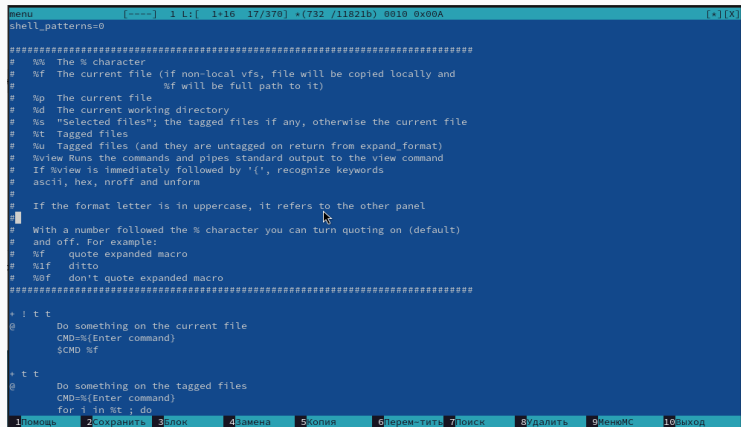
Рис. 16: Переход в домашний каталог



```
mc.ext.ini [----] 0 L: [ 1+ 0 1/1136] *(0 /26909b) 0035 0x023 [*][X]
# Midnight Commander 4.0 extension file

# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located as before as after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# If there are more than one sections with the same name in this file, the first
# section will be used.
#
# [Default] should be a catch-all action and come last.
```

Рис. 17: Просмотр файла расширений

A screenshot of the Midnight Commander file manager's menu system. The window title is 'menu' with standard window controls. The menu is open, showing a list of options. At the top, it says 'shell_patterns=0'. The menu items include: '%%' (The % character), '%f' (The current file), '%p' (The current file), '%d' (The current working directory), '%s' (Selected files), '%t' (Tagged files), '%u' (Tagged files), '%view' (Runs the commands), and a section for format letters (ascii, hex, nroff, uniform). Below this, there are two sections for macros: one for the current file (CMD=%{Enter command}) and one for tagged files (CMD=%{Enter command}). At the bottom, a status bar shows numbered shortcuts: 1.Помощь, 2.Сохранить, 3.Блок, 4.Замена, 5.Копия, 6.Перем-тить, 7.Поиск, 8.Удалить, 9.МенюМС, 10.Выход.

```
menu [=====] 1 L: [ 1*16 17/370] *(732 /11821b) 0010 0x00A [ * ] [ X ]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#                               %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, nroff and uniform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
#   %f   quote expanded macro
#   %!f  ditto
#   %0f  don't quote expanded macro
#####

+ ! t t
@      Do something on the current file
      CMD=%{Enter command}
      $CMD %f

+ t t
@      Do something on the tagged files
      CMD=%{Enter command}
      for i in %t ; do

1.Помощь  2.Сохранить  3.Блок  4.Замена  5.Копия  6.Перем-тить  7.Поиск  8.Удалить  9.МенюМС  10.Выход
```

Рис. 18: Просмотр файла меню

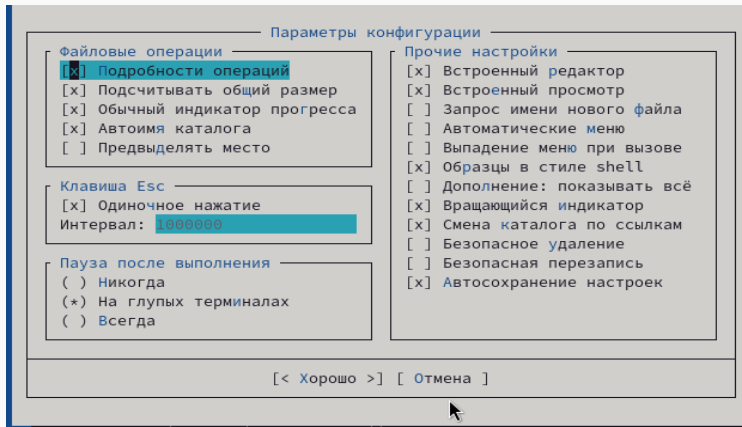


Рис. 19: Конфигурация

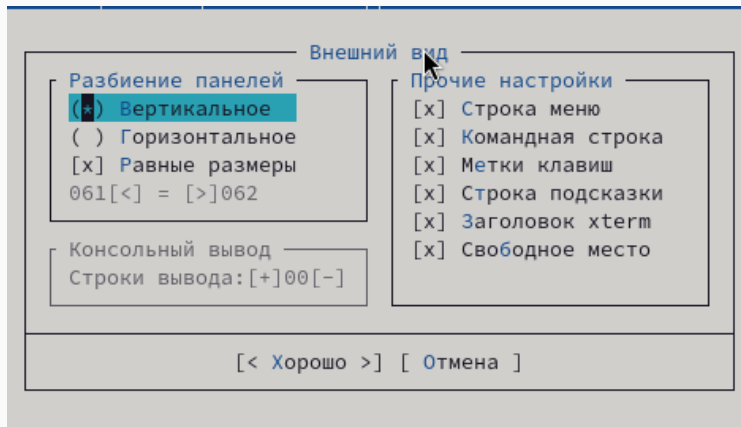


Рис. 20: Внешний вид

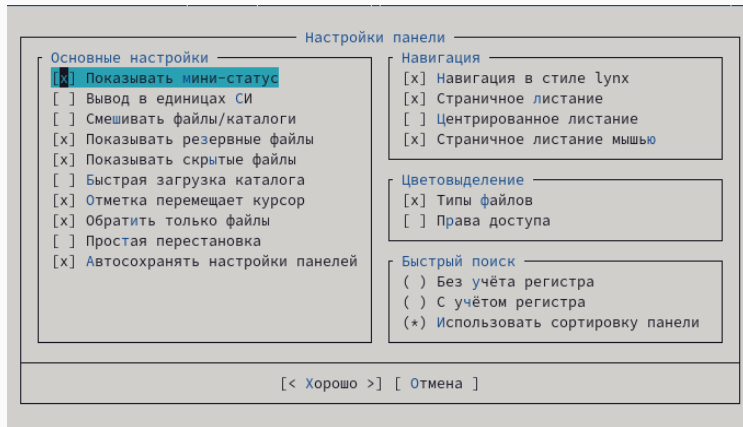


Рис. 21: Настройки панелей

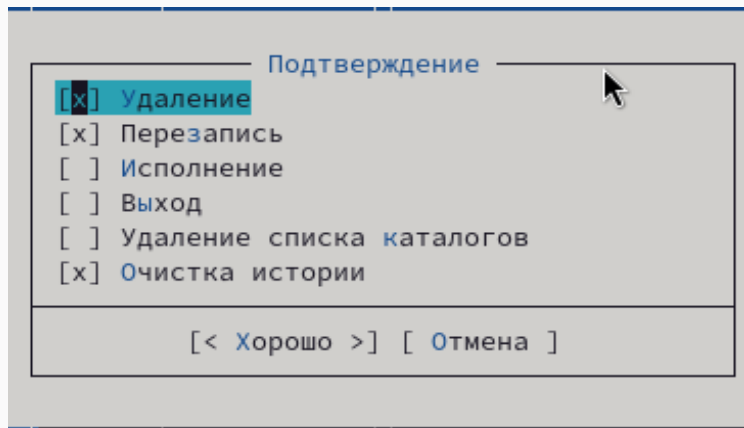


Рис. 22: Подтверждение

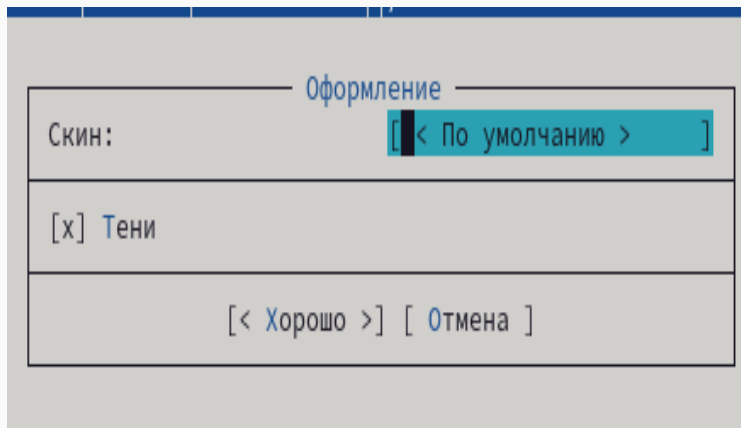


Рис. 23: Оформление

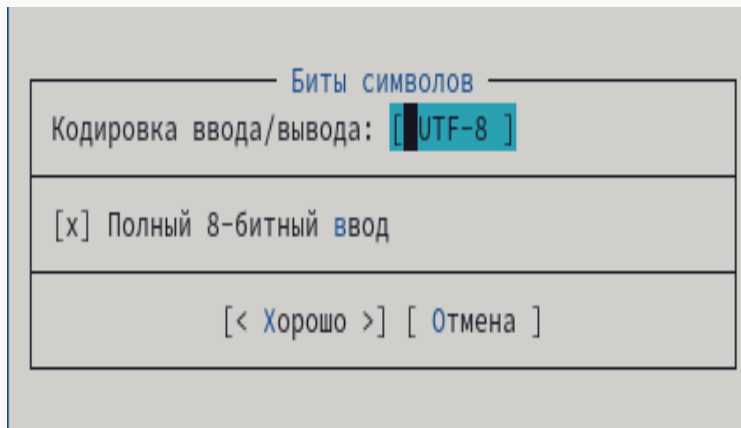


Рис. 24: Кодировка символов

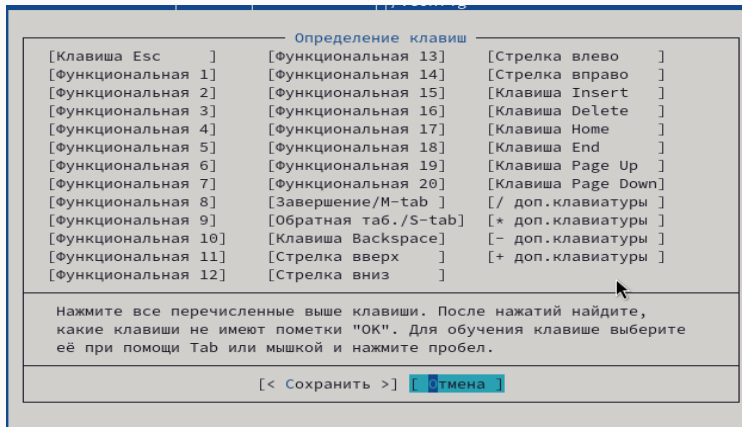
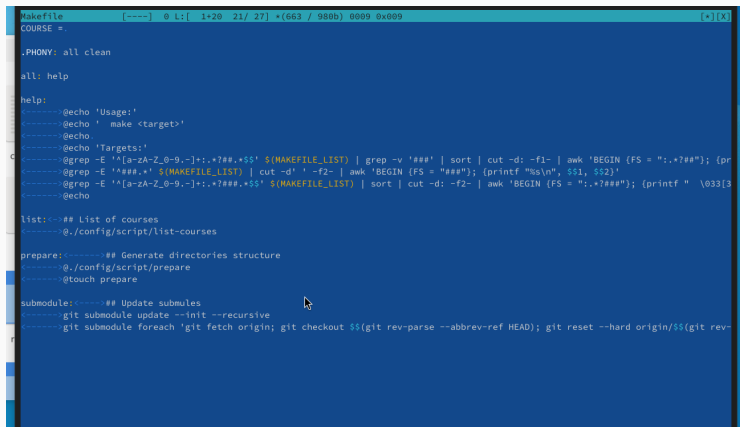


Рис. 25: Распознавание клавиш

Работа с редактором Midnight Commander



```
Makefile [----] 0 L: [ 1+20 21/ 27] *(663 / 988b) 0009 0x009 [*][X]
COURSE = .

.PHONY: all clean

all: help

help:
<-----@echo 'Usage:'
<-----@echo ' make <target>'
<-----@echo.
<-----@echo 'Targets:'
<-----@grep -E '[a-zA-Z_0-9-]+.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
<-----@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
<-----@grep -E '[a-zA-Z_0-9-]+.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf " \033[3
<-----@echo

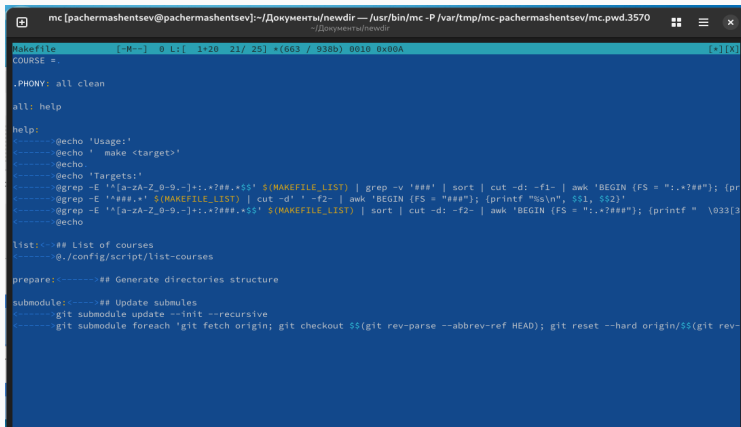
list:<--## List of courses
<-----@./config/script/list-courses

prepare:<-----## Generate directories structure
<-----@./config/script/prepare
<-----@touch prepare

submodule:<-----## Update submules
<-----git submodule update --init --recursive
<-----git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
```

Рис. 26: Файл с текстом

Работа с редактором Midnight Commander



```
mc [pachermashentsev@pachermashentsev]:~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~/Документы/newdir
Makefile      [-M--]  0 L: [ 1+20  21/ 25] *(663 / 938b) 0010 0x00A      [*][X]
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $1, $2}'
<----->@grep -E '[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf "  \033[3
<----->@echo

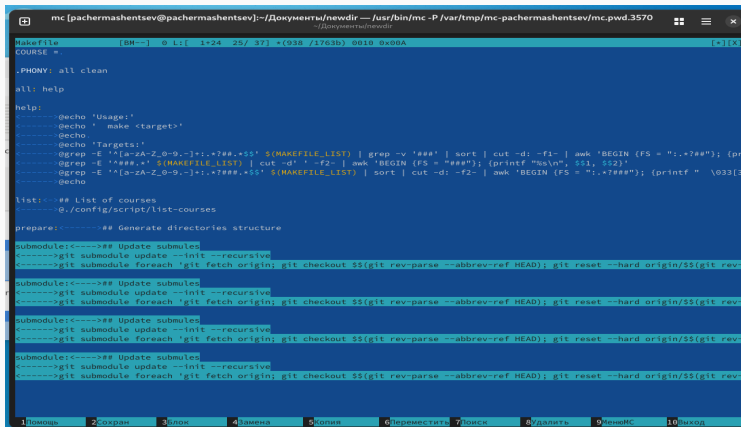
list:<-->## List of courses
<----->@./config/script/list-courses

prepare:<----->## Generate directories structure

submodule:<----->## Update submodules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
```

Рис. 27: Файл с текстом

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The top bar displays the user 'mc [pachermashentsev@pachermashentsev]:~/Документы/newdir' and the command line '/usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570'. The main window shows a file named 'Makefile' with a blue background. The file content is visible, including comments and shell commands. A terminal window is open at the bottom, showing the output of the 'list' command, which lists the contents of the 'courses' directory. The terminal output shows a list of courses with their respective paths and names.

```
mc [pachermashentsev@pachermashentsev]:~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~/Документы/newdir
Makefile [BH--] 0 Li[ 1+24 25/ 37] *(938 /1763b) 0810 0x00A [*][X]
COURSE =.
.PHONY: all clean
all: help
help:
  @echo 'Usage:'
  @echo '  make <target>'
  @echo
  @echo 'Targets:'
  @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
  @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {printf "%s\n", $
  @echo
list:## List of courses
  @./config/script/list-courses
prepare:## Generate directories structure
submodule:## Update submodules
  @git submodule update --init --recursive
  @git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
submodule:## Update submodules
  @git submodule update --init --recursive
  @git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
submodule:## Update submodules
  @git submodule update --init --recursive
  @git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
submodule:## Update submodules
  @git submodule update --init --recursive
  @git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
```

Рис. 28: Копирование фрагмента

Работа с редактором Midnight Commander

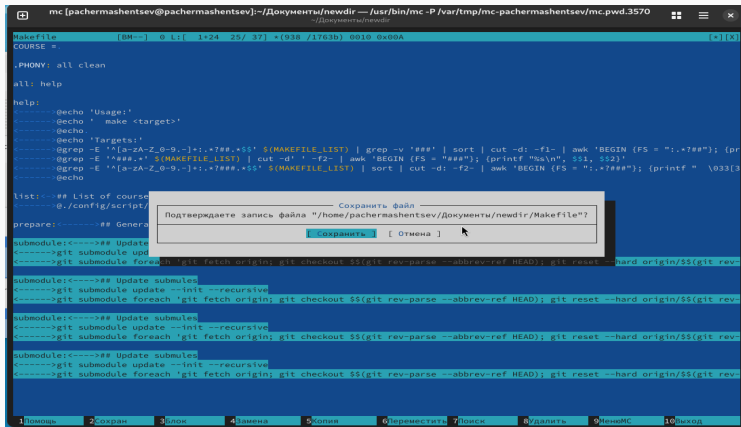


Рис. 29: Сохранение

Работа с редактором Midnight Commander

```
mc [pachermashentsev@pachermashentsev]~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~Документы/newdir

Makefile  [BM~]  0 Li:  1+24  25/ 29] * (93b /1213b) 0010 0x00A  [*] [X]
COURSE =

.PHONY: all clean

all: help

help:
#####@echo 'Usage:'
#####@echo ' make <target>'
#####@echo
#####@echo 'Targets:'
#####@grep -E '^[a-zA-Z_0-9-~]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
#####@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
#####@grep -E '^[a-zA-Z_0-9-~]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf " \033[3
#####@echo

list:##### List of courses
#####@./config/script/list-courses

prepare:##### ## Generate directories structure

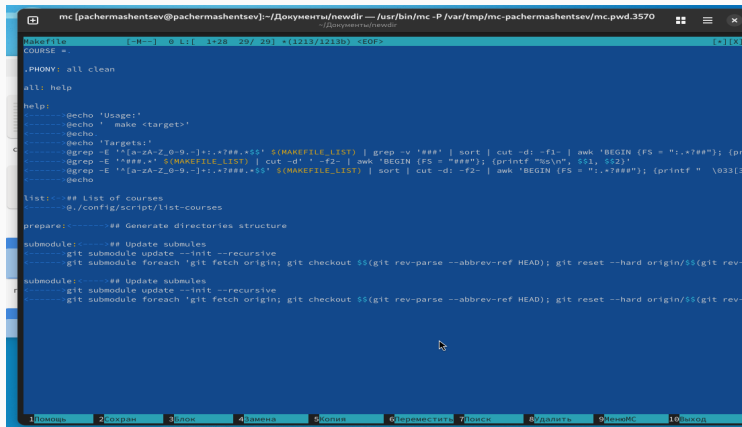
submodule:##### ## Update submodules
#####git submodule update --init --recursive
#####git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
#####

submodule:##### ## Update submodules
#####git submodule update --init --recursive
#####git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
#####
```

1 Помощь 2 Сохран 3 Блок 4 Замена 5 Скопир 6 Переместить 7 Поиск 8 Удалить 9 Переименовать 10 Выход

Рис. 30: Отмена

Работа с редактором Midnight Commander



```
mc [pachermashentsev@pachermashentsev]:~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~/Документы/newdir

Makefile  [-M--]  0  L:  1+28  29/ 29  *(1213/1213b) <EOF>  [*][X]
COURSE =.

.PHONY: all clean

all: help

help:
-----@echo 'Usage:'
-----@echo ' make <target>'
-----@echo
-----@echo 'Targets:'
-----@grep -E '[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
-----@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
-----@grep -E '[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf " \033[3
-----@echo

list:<--## List of courses
----->@./config/script/list-courses

prepare:<-----## Generate directories structure

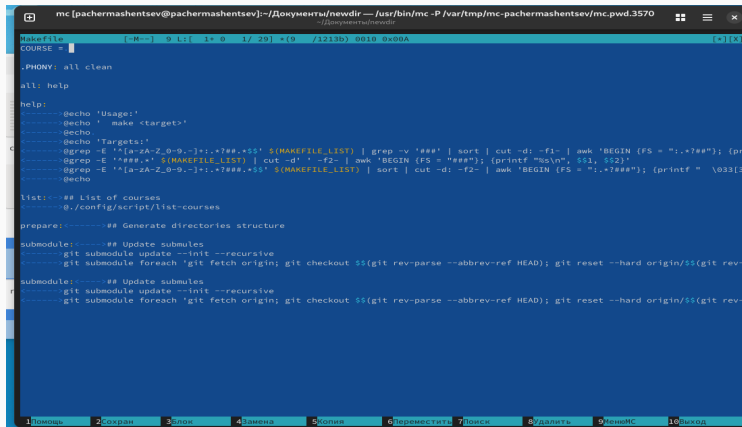
submodule:<-----## Update submules
-----git submodule update --init --recursive
-----git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-

submodule:<-----## Update submules
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1Помощь 2Сохран 3Блок 4Замена 5Копия 6Переместить 7Поиск 8Удалить 9НенаМС 10Выход
```

Рис. 31: Переход в конец файла

Работа с редактором Midnight Commander



```
mc [pachermashentsev@pachermashentsev]:~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~/Документы/newdir
Makefile  [-M--]  9 L: [ 1= 0 1/ 29] * (0 /1213b) 0010 0x00A [*] [X]
COURSE =
.PHONY: all clean
all: help
help:
-----@echo 'Usage:'
-----@echo ' make <target>'
-----@echo
-----@echo 'Targets:'
-----@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN {FS = ".*?##"}; {pr
-----@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
-----@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?##"}; {printf " \033[3
-----@echo

list:---## List of courses
-----@./config/script/list-courses

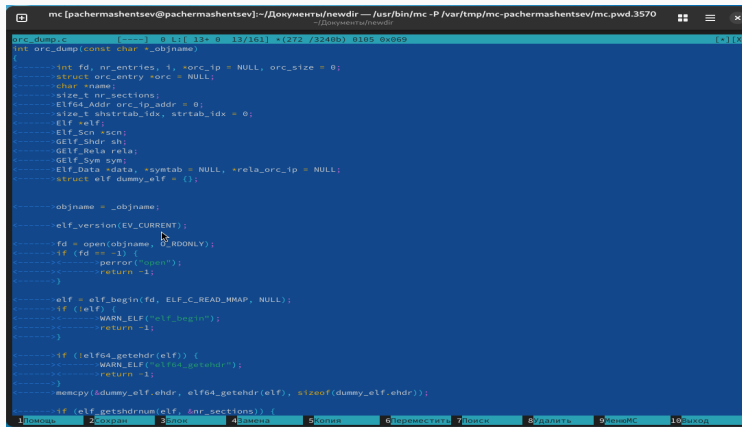
prepare:-----## Generate directories structure

submodule:-----## Update submodules
-----git submodule update --init --recursive
-----git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-

submodule:-----## Update submodules
-----git submodule update --init --recursive
-----git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --hard origin/$(git rev-
```

Рис. 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [pachermashentsev@pachermashentsev]:~/Документы/newdir — /usr/bin/mc -P /var/tmp/mc-pachermashentsev/mc.pwd.3570
~/Документы/newdir
orc_dump.c  [-----]  0 L: [ 13+ 0 13/161 ] * (272 / 3240b) 0105 0x069
int orc_dump(const char *objname)
{
    >int fd, nr_entries, i, *orc_ip = NULL, orc_size = 0;
    >struct orc_entry *orc = NULL;
    >char *name;
    >size_t nr_sections;
    >Elf64_Addr orc_ip_addr = 0;
    >size_t shstrtab_idx, strtab_idx = 0;
    >Elf *elf;
    >Elf_Scn *scn;
    >GElf_Shdr sh;
    >GElf_Rela rela;
    >GElf_Sym sym;
    >Elf_Data *data, *syntab = NULL, *rela_orc_ip = NULL;
    >struct elf dummy_elf = {};

    >objname = _objname;
    >elf_version(EV_CURRENT);

    >fd = open(objname, O_RDONLY);
    >if (fd == -1) {
    >>error("open");
    >>return -1;
    >}

    >elf = elf_begin(fd, ELF_C_READ_MMAP, NULL);
    >if (!elf) {
    >>WARN_ELF("elf_begin");
    >>return -1;
    >}

    >if (!elf64_getehdr(elf)) {
    >>WARN_ELF("elf64_getehdr");
    >>return -1;
    >}
    >memcpy(&dummy_elf.ehdr, elf64_getehdr(elf), sizeof(dummy_elf.ehdr));

    >if (elf_getshdrnum(elf, &nr_sections)) {
```

Рис. 33: Файл с программой

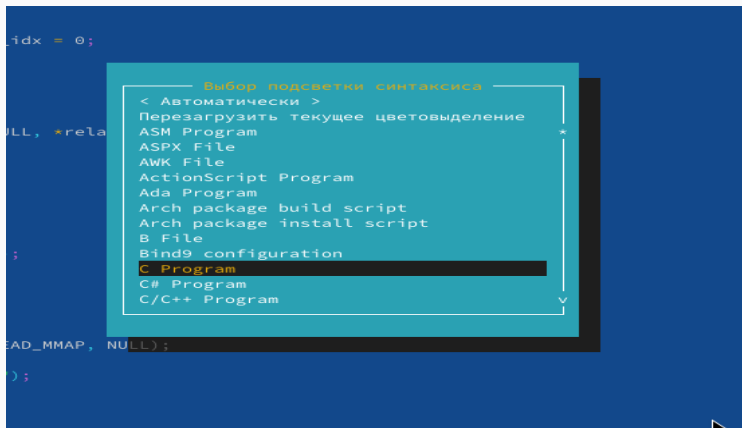


Рис. 34: Цветовыделение синтаксиса

Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.